Privacy-preserving techniques using zero knowledge proof in public Ethereum

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ELCA Informatique SA

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Agenda

1 — Why Privacy is important in blockchain ?

2 — Theory and concept around Zero Knowledge proof

3 — Application of zk-SNARKs in public Ethereum Blockchain

4 — Q & A

Importance of privacy

- Essential human right
- Allow people exercise unique moral choices
- Enable our societies evolved around is healthy, and makes it more resilient and more humane

"The right to personal privacy is precious. Without it, we are all potential victims for a prying secret police."

—Lewis B. Smedes

Why privacy matters

Once data is embedded in the blockchain it cannot be altered

Right to be forgotten

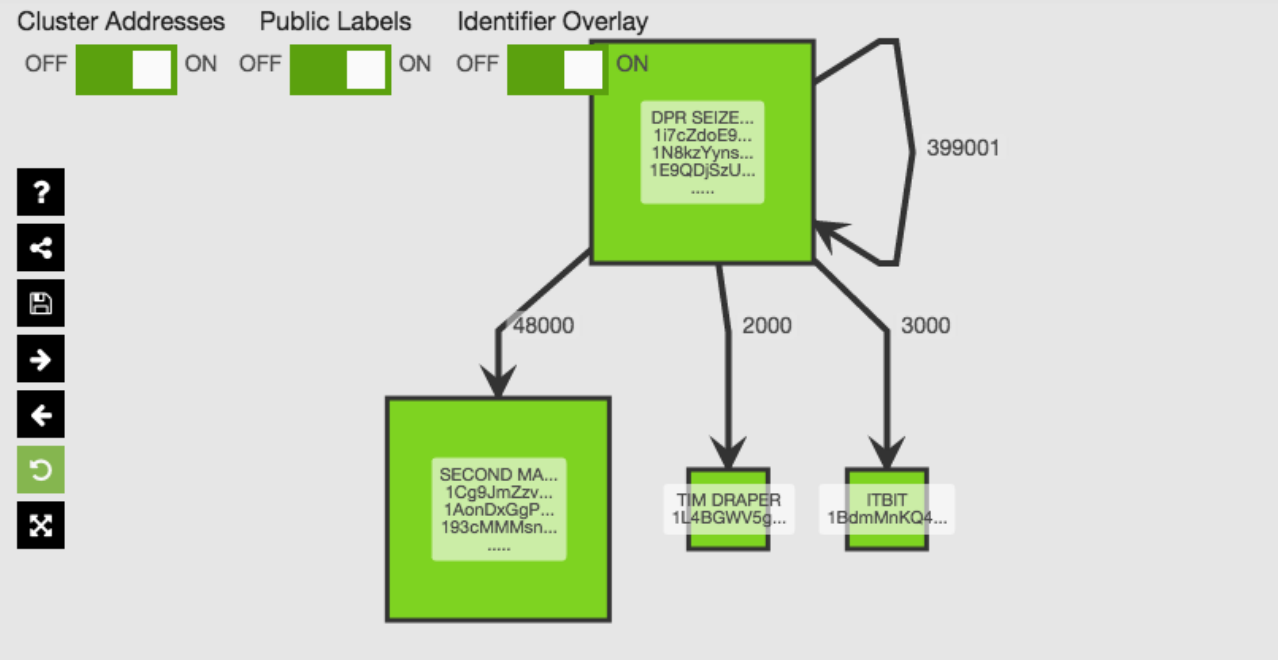
Right to change personal data

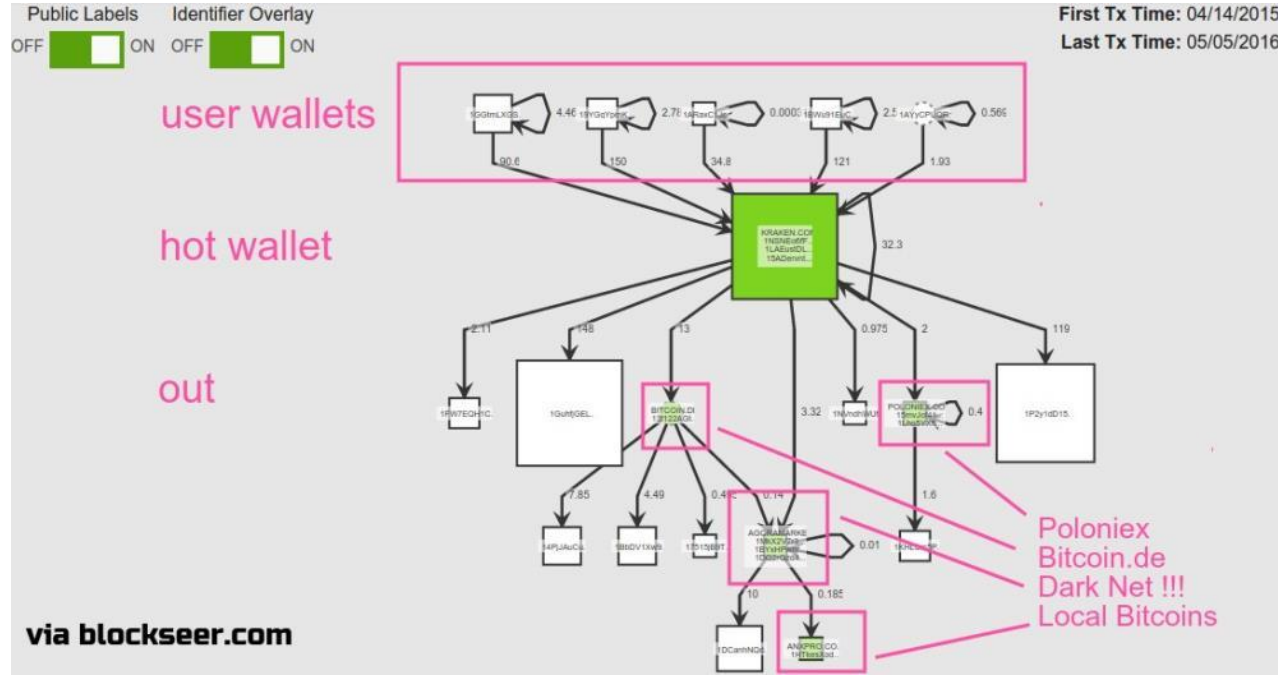
Right to hide personal info

Do you think Bitcoin is still anonymous ?

- **Wallet address** re-use, linking your transactions together into a single profile.
- **IP address** re-use.
- **Combining inputs** from multiple transactions,
- **Using mobile wallet**, effectively revealing to a third party your full set of addresses.

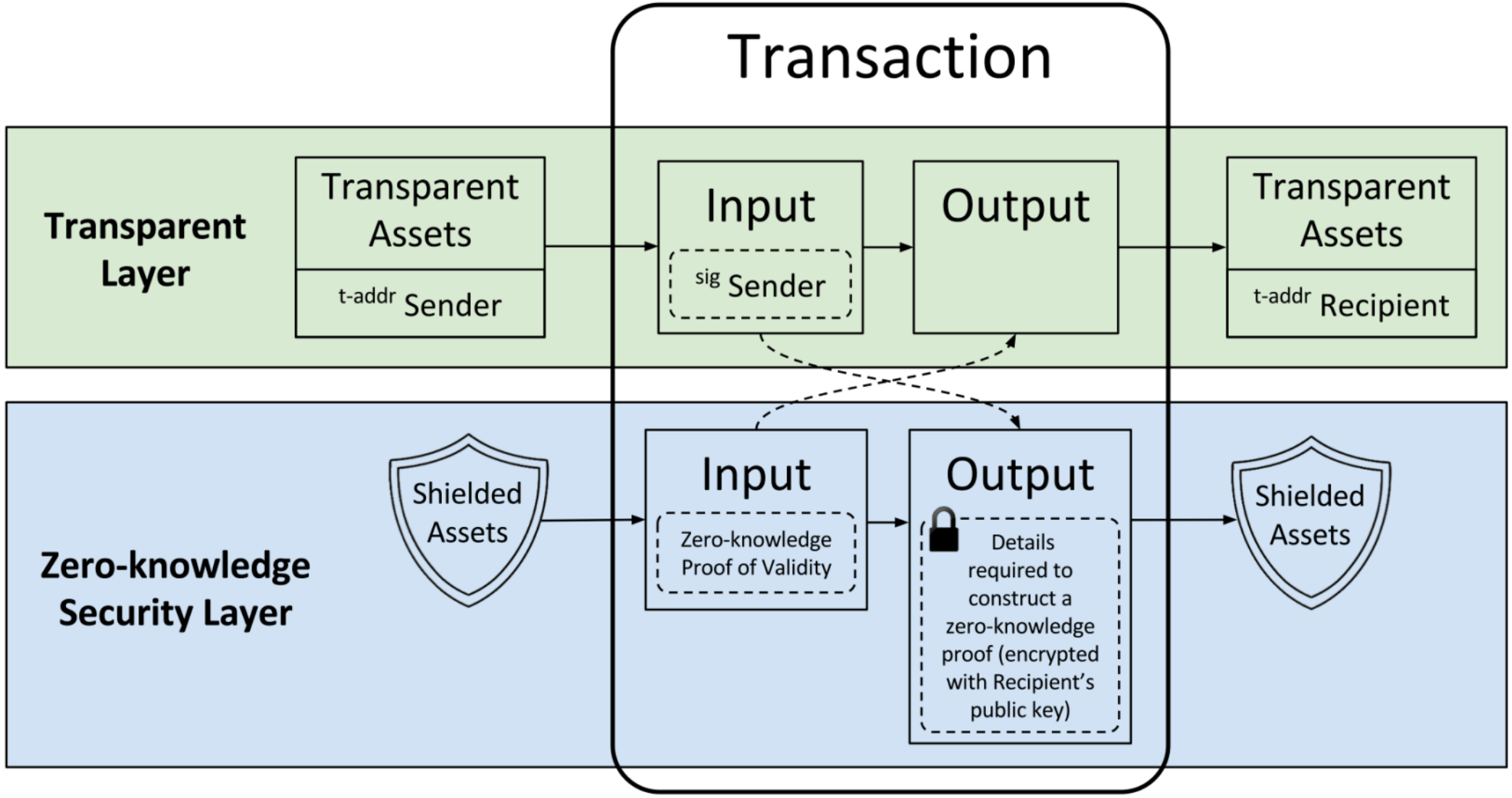
Following a bitcoin transaction



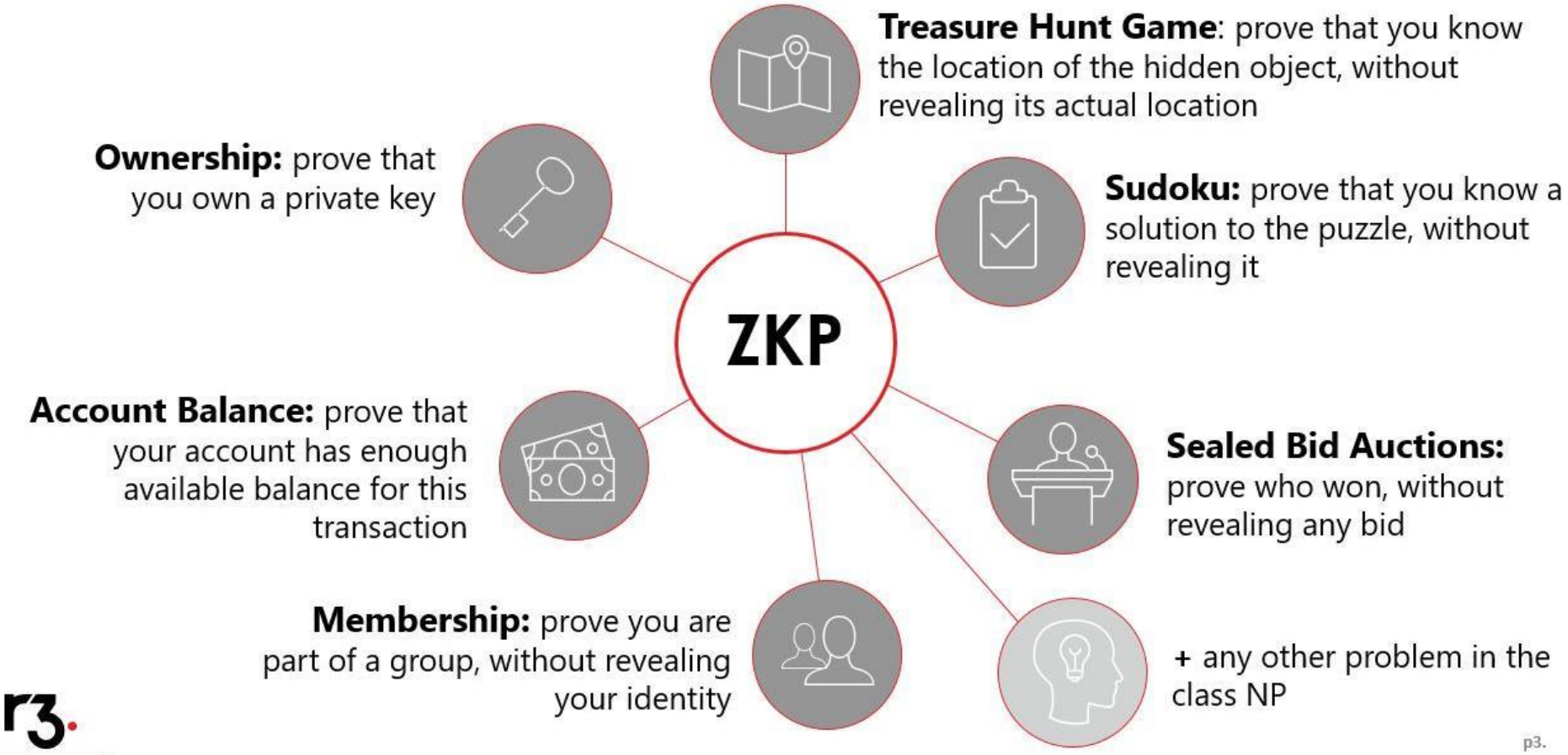


CASH

How to make private transaction in the blockchain



Application of Zero Knowledge Proof



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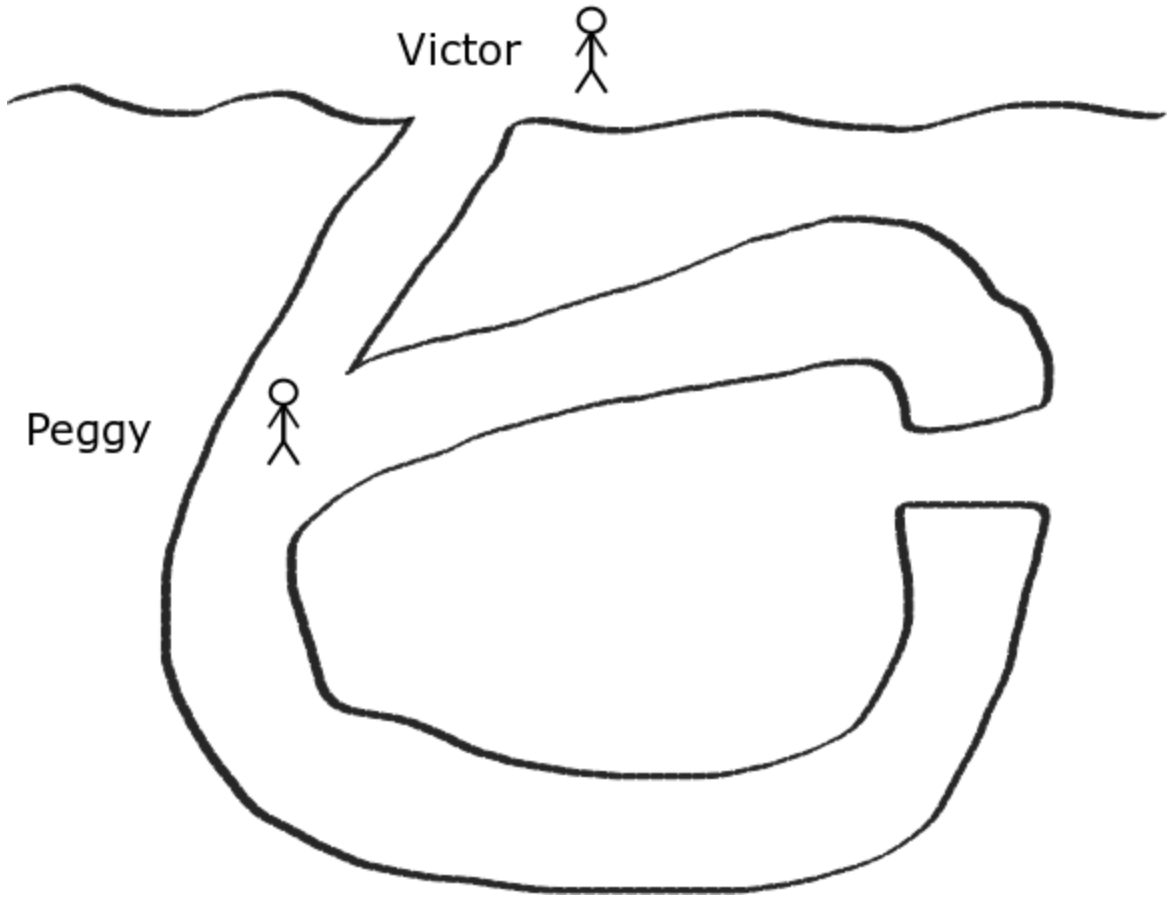


Zero-knowledge proof

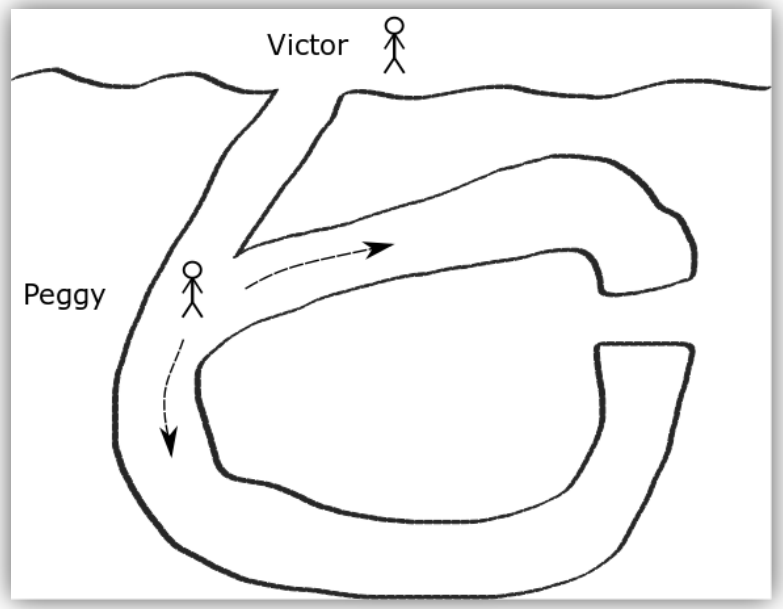
In a Zero-knowledge (ZK) proof we have 2 participants:

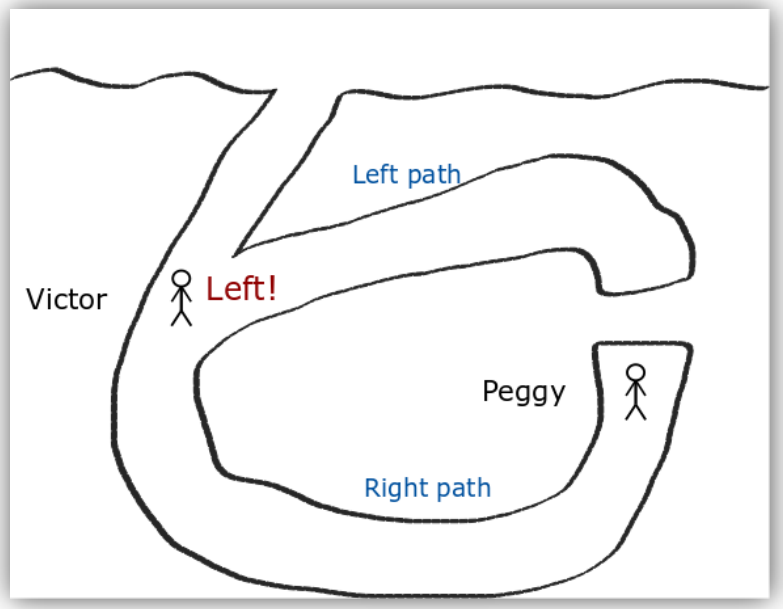
- **Prover - Peggy**
- **Verifier - Victor**

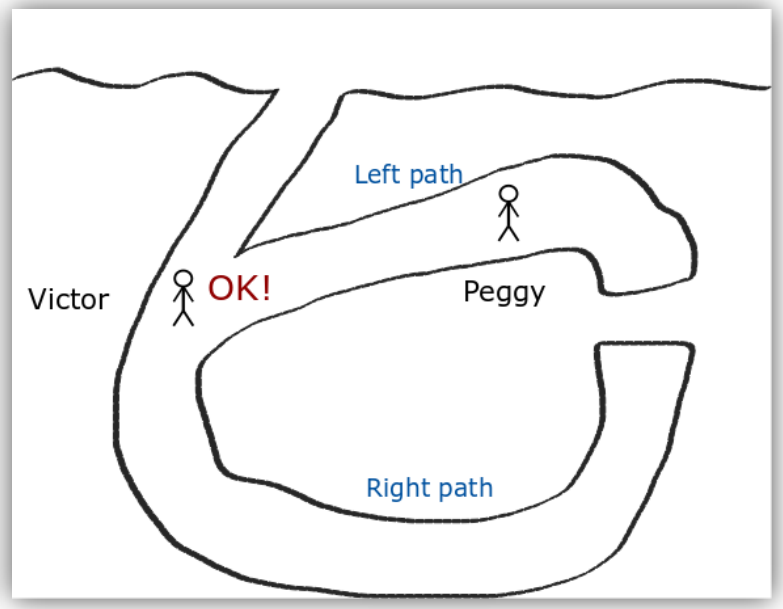
The prover wants to convince the verifier that a statement is true without leaking any extra information.

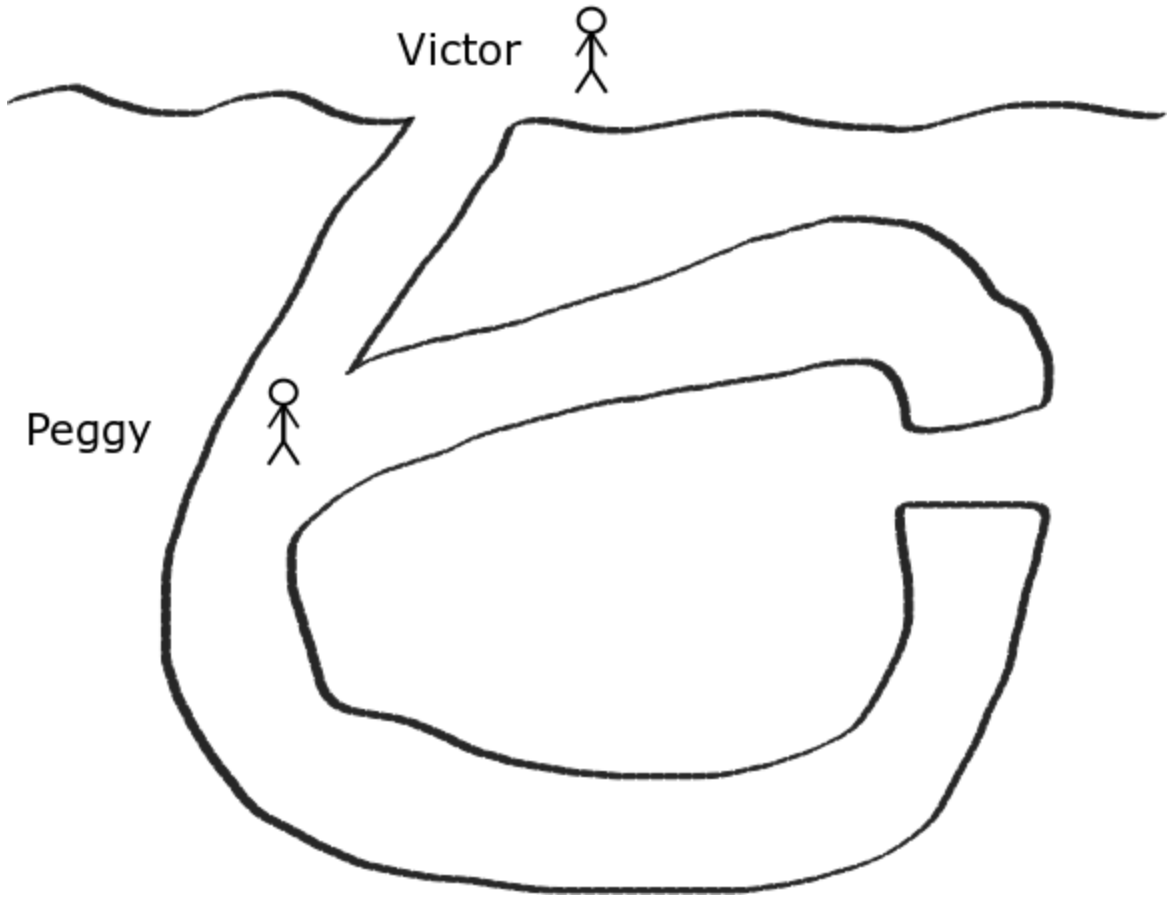
E.g. the prover wants to convince the verifier that he has a secret without revealing it.

Zero-knowledge proof









Zero-knowledge proof

Properties of ZK proofs:

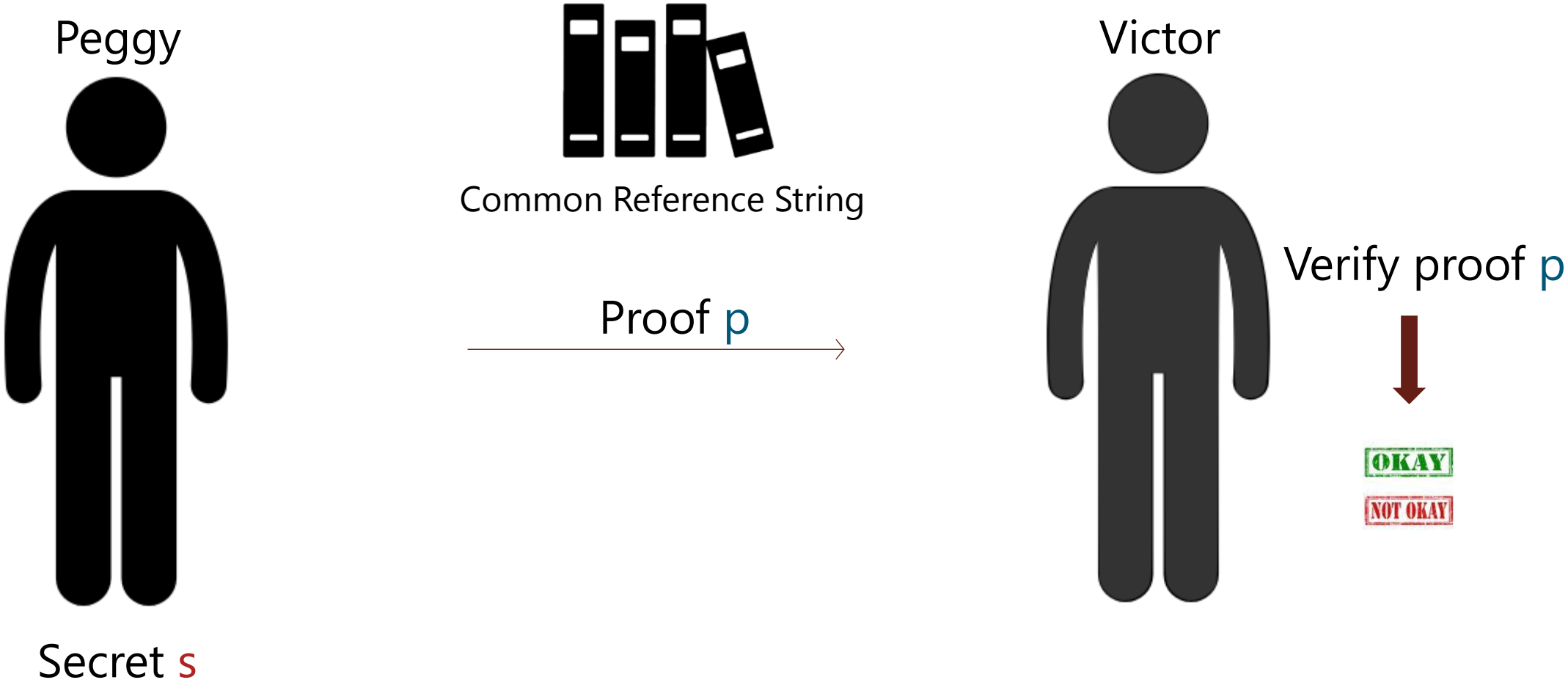
- **Termination:** the protocol finishes after a number of steps.
- **Completeness:** if the statement is true then the verifier will be convinced of this at the end of the protocol.
- **Soundness:** if the statement is false then it is unlikely that the verifier will be fooled.
- **Zero-knowledge:** at the end of the protocol no sensitive information is leaked to the verifier.

zkSNARK

Zero-knowledge Succinct Non-interactive ARguments of Knowledge

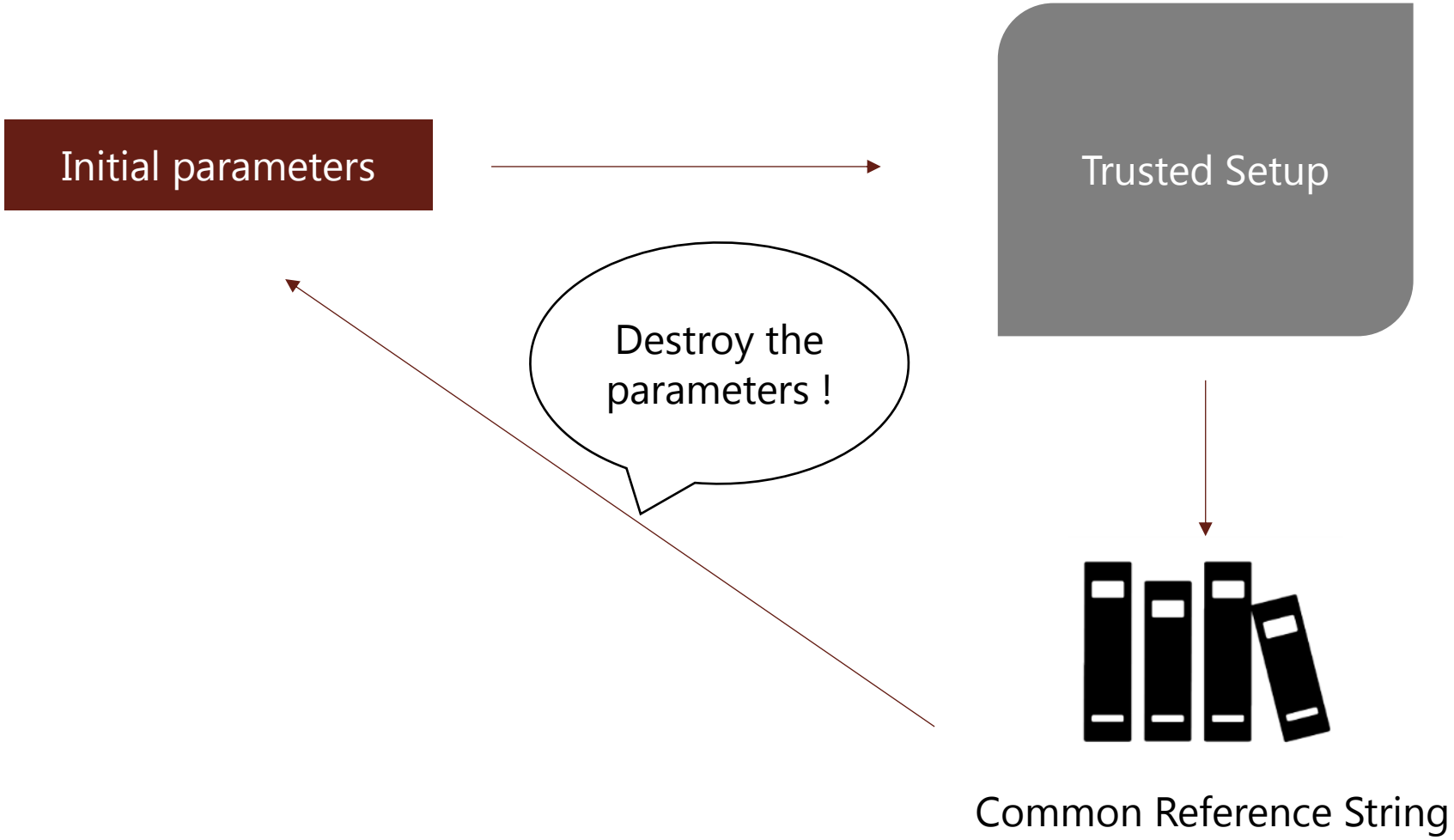
- Zero-knowledge: proving existence of a secret without revealing it
- Succinct: the communication volume is small
- Non-interactive: no interaction
- Arguments of Knowledge: proving that one knows the secret

Non-interactive Zero-knowledge proof (NIZK)



Source: https://thenounproject.com

Common Reference String (CRS)

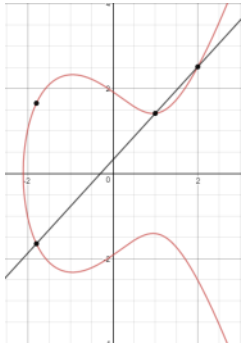


Pairing on elliptic curves

Pairing: mathematical tool used to instantiate a NIZK

Elliptic curve:

$$E_{a,b} = \{O\} \cup \{(x, y); y^2 = x^3 + ax + b\}$$

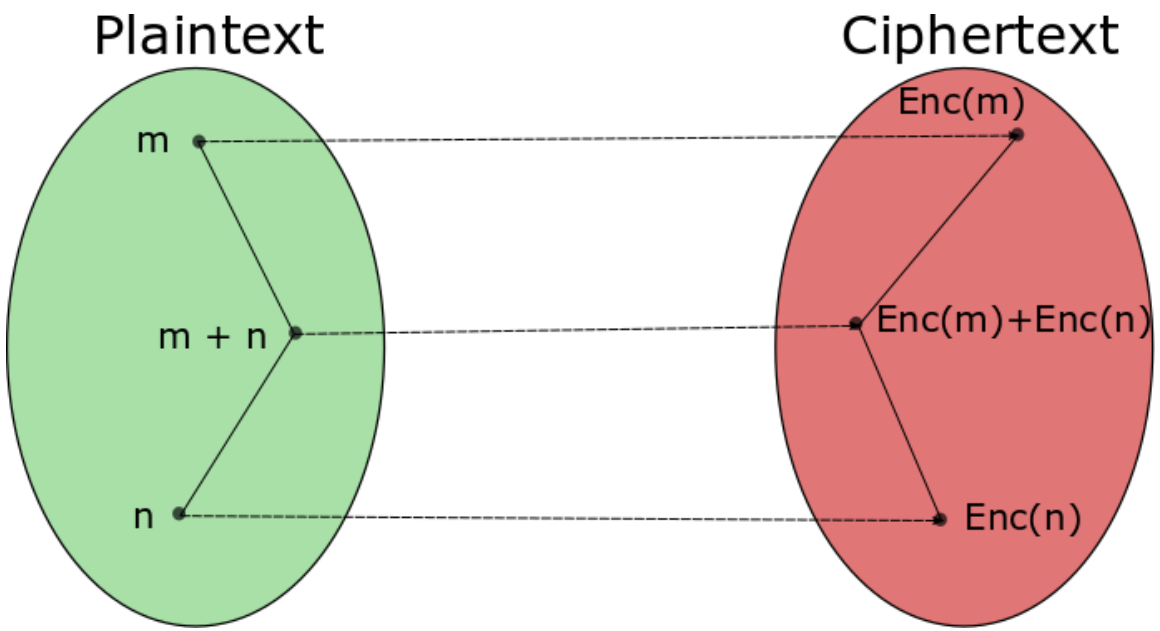


Efficiency

RSA (bit length)	**EC** (bit length)	Security level
1248	160	80
2048	224	112

Used by

Homomorphic evaluation/ encryption



$$Enc(m) + Enc(n) = Enc(m + n)$$

Homomorphic encryption allows to perform operations on sensitive data while keeping it encrypted.

Use case: secure cloud computing

Agenda

1 — Privacy in Ethereum

2 — Byzantium

3 — Zk-SNARKs

4 — Demo

5 — ZoKrates

6 — Conclusions

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Ethereum: public but private but public…

Byzantium fork

The current incarnation of Geth contains all the Byzantium EIPs implemented and also features the fork block number **1,700,000** for the Ropsten testnet transition. The block numbers for Rinkeby and the main Ethereum network will be finalized when Ropsten is deemed stable.

You can find details about individual protocol updates at the following locations:

- EIP 98: Removal of intermediate state roots from receipts (#14750).
- Expanded by EIP 658: Embedding transaction return data in receipts (#15014).
- EIP 100: Change difficulty adjustment to target mean block time including uncles (#14733).
- EIP 198, EIP 212 (197) and EIP 213 (196): Precompiled contracts for modular exponentiation; elliptic curve addition, scalar multiplication and pairing (#14959, #14993).
- EIP 214 (116): Expanding the EVM with static contract calls (#14978).
- EIP 211: Expanding the EVM with dynamically sized return data (#14981).
- EIP 206 (140): Expanding the EVM with cheap state revertals (#14983).
- EIP 649: Delaying the difficulty bomb and reducing the block reward (#15028).
- EIP 684: Preventing overwriting contracts (Byzantium prep) (#15039).

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Byzantium what?

zk-SNARKs what?

Secure | https://www.reddit.com/r/ethereum/comments/712idt/ethereum_testnet_just_verified_a_zcash_transaction/

Apps | Elca | Training | Technical Articles | Blockchain | Development | Personal | Summercamp Schedu | Noisli - Improve Focu

ELI2weeksintocrypto

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[–] **Der_Bergmann** **10 points** 25 days ago

the transaction shows sender and receiver. What's the infos that is hided by the zero knowledge proof?

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[–] **chriseth** Ethereum - Christian Reitwießner [S] **44 points** 25 days ago

What is verified here is the zkSNARK part of some real transaction on the real zCash network. I pulled the data out of the chain some time around January, I think. I have no idea who created the transaction or how much money is sent inside it, but now the Ethereum network knows it is correct :-)

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[–] **Der_Bergmann** **6 points** 25 days ago

Thanks for the great explanation.

I assume the plan is to let Ethereum transactions happen this way, that everybody knows, it is correct, but nobody know what's happening inside ...

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[–] **chechuch** **2 points** 25 days ago

Great! And Ty from for the explanation! Good job!!!

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[–] **legalgrayarea** **4 points** 25 days ago

It's not currently using the secure address format it seems. It's simply a test of zsnak verification.

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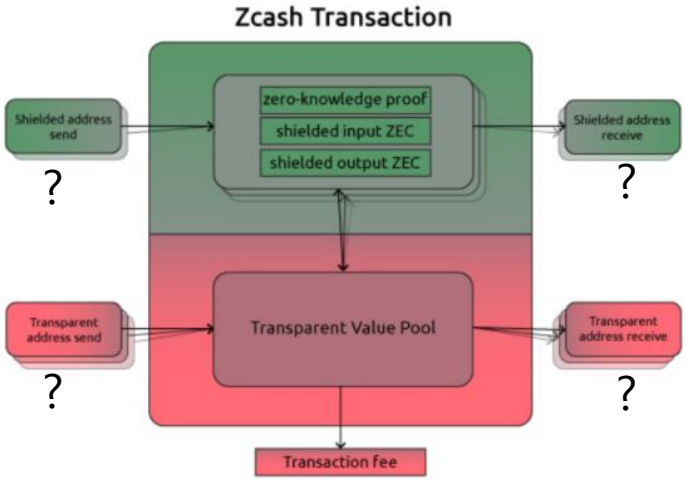
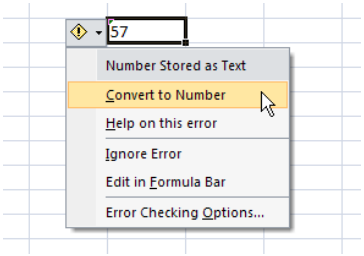
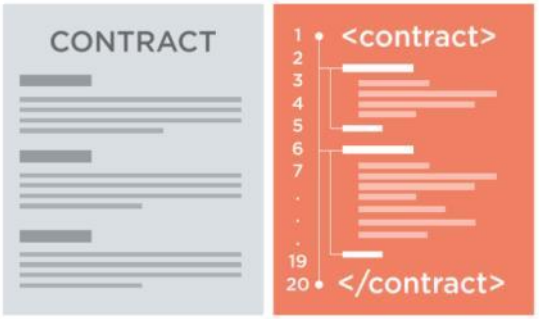
How does it work?



A high-level view of a Zcash transaction



TEST Contract



Library: Pairing

Pre-compiled EC operations

Verified by ETHEREUM

Reference: https://www.reddit.com/r/ethereum/comments/712idt/ethereum_testnet_just_verified_a_zcash_transaction/

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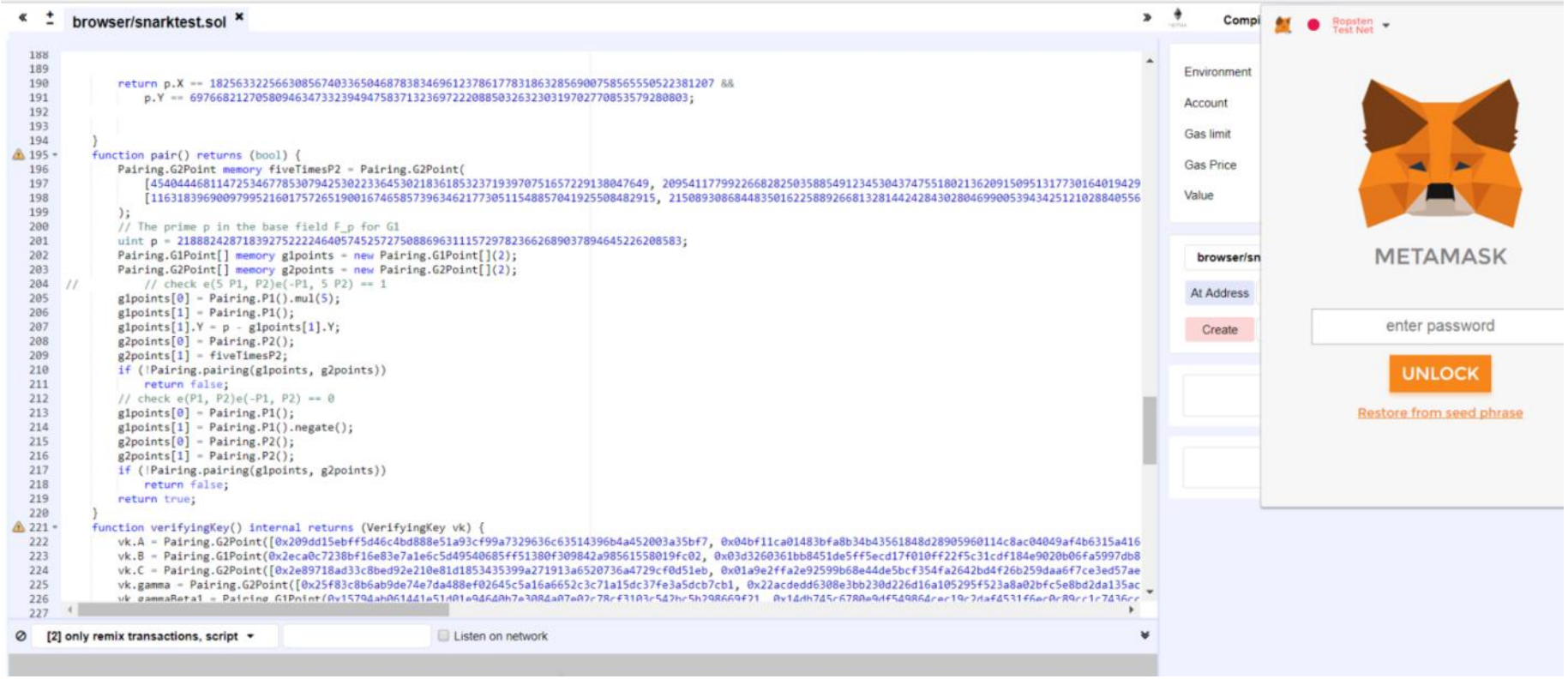
Demo…


browser/snarktest.sol
Compi
Ropsten Test Net
Environment
Account
Gas limit
Gas Price
Value
browser/sn
At Address
Create
METAMASK
enter password
UNLOCK
Restore from seed phrase
[2] only remix transactions, script
Listen on network

ZoKrates… what?

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ZoKrates

"… ZoKrates allows information to be obscured off the main ethereum blockchain and then uploaded into a smart contract the network can still verify, all without exposing contract information."

It is another type of ethereum smart contract. A custom version of the self-executing code that runs on top of the network, the tool serves as a way to transfer a zk-SNARKs operation onto the blockchain and to verify that that information is valid.

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Conclusions

- Privacy is the holy-grail of public blockchains
- zk-SNARKs allows to verify something without knowing the details
- Byzantium fork on Ethereum brings new functionality to facilitate the verifiability of zk-SNARKS
- zk-SNARKs is difficult to grasp and even more difficult to code and automate
- More tooling is needed and ZoKrates attempts to fill the gap

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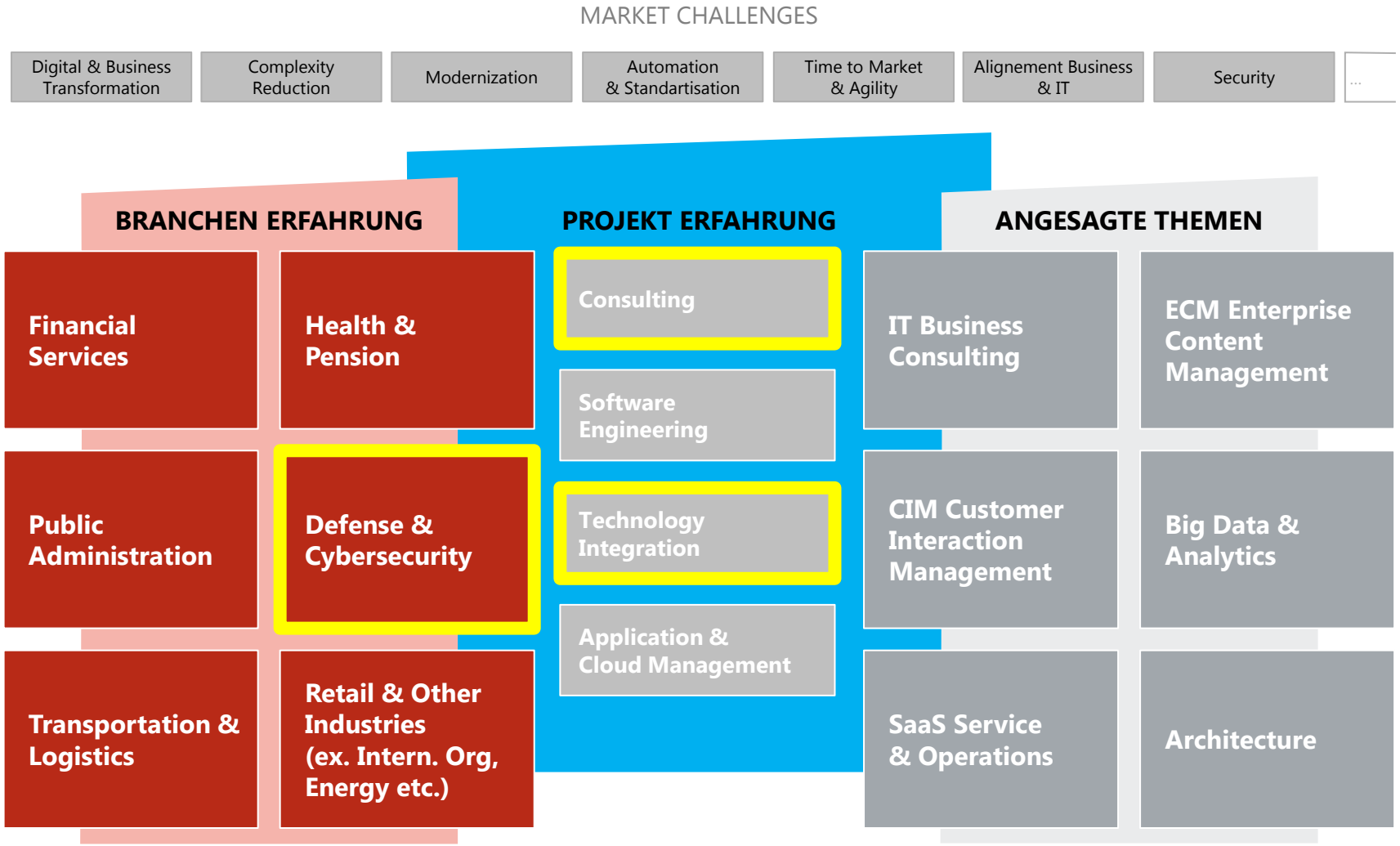
Innovations @ ELCA

ELCA is working on innovation projects which aim to cover the following domains:

- Cloud Security and Searchable Encryption in the cloud
- ChatBot and Natural Language processing
- BigData
- Blockchain

ELCA expertise



Blockchain team

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Business expert
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Industry
Business expert
Financial

Dev Team
Smart contract
> 10 engineers

Head of Division / Senior Consultant
Blockchain evangelist
> 10 senior manager

Blockchain Team

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Ethereum Expert
Hyperledger
Smart Contract
Cloud Architect

Reto F.
Security Architect
Blockchain Expert
Zurich

Industry
Business expert
Public sector

Laurent P
Blockchain Java
Hyperledger
Smart Contract
Architect

Sébastien P
Lead Architect

Industry
Business expert
Healthcare

Dedicated skilled Blockchain experts

How ELCA can help on Blockchain technology

Blockchain Hands-on

- Discuss Blockchain technology
- Explore customer business model
- Show Blockchain Application demo

Experience Blockchain

- Understand Blockchain concepts & elements
- Hands on with Blockchain on Ethereum / Hyperledger / Others
- Standard demo customization

Provide Expertise and First Project

- Define business challenge
- Agile iterations incrementally build project functionality
- Enterprise integration
- Provide Blockchain Dev expert and consulting services

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